

A Hybrid Deep Learning Framework for Automated Glaucoma Detection and Severity Estimation Using Fundus Images with Explainable Artificial Intelligence

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ABSTRACT

Glaucoma silently damages the optic nerve over years, and by the time most patients notice symptoms, a significant portion of their vision is already lost. Early, automated screening is therefore not just useful but necessary. Existing automated systems largely treat glaucoma detection as a binary classification problem, leaving clinicians without the quantitative severity information they need to plan treatment. This paper proposes a unified deep learning framework that simultaneously classifies glaucoma from retinal fundus images and estimates disease severity using the cup-to-disc ratio (CDR), all within a single end-to-end pipeline. EfficientNet-B3 was fine-tuned on the REFUGE2 dataset (2,000 images) with Focal Loss for binary classification, while a hybrid ResNet50–Vision Transformer (ViT) architecture was developed for CDR regression by fusing local convolutional features with global attention representations. Explainability is built in from the start: Grad-CAM highlights the fundus regions driving classification decisions, while attention rollout maps the transformer’s spatial focus during CDR estimation. Evaluation on two independent public datasets shows that EfficientNet-B3 achieves ROC-AUC of 0.841 (95% CI: 0.787–0.890), sensitivity of 0.738, and specificity of 0.813 on the REFUGE2 held-out test set ($n = 400$). On DRISHTI-GS, CDR regression via five-fold cross-validation achieved Pearson $r = 0.673$ (95% CI: 0.485–0.801) with MAE = 0.107. An ablation study on DRISHTI-GS confirms that the hybrid architecture outperforms a ResNet-50 baseline by 6.6 percentage points in AUC. Severity grading across four clinical categories reaches 92.0% within-one-category accuracy. Grad-CAM outputs consistently highlight the optic disc and neuroretinal rim, confirming clinically meaningful model behaviour. The proposed system offers a practical, transparent, and end-to-end solution for glaucoma screening in resource-limited settings where specialist access is scarce.

1. Introduction

Glaucoma is one of the most prevalent causes of permanent blindness worldwide. Unlike many other eye diseases, it develops gradually and with almost no noticeable symptoms in the early stages. The World Health Organisation estimates that around 80 million people currently live with the condition and this number is expected to reach 110 million by 2040 [1]. The tragedy of glaucoma lies not just in its prevalence but in its irreversibility: once the optic nerve is damaged, the lost vision cannot be restored. Early detection is therefore the only real intervention point available to clinicians.

Retinal fundus photography provides a non-invasive and cost-effective window into the optic nerve. Among the structural indicators evaluated in these images, the cup-to-disc ratio (CDR) stands out as the most widely used marker of glaucoma progression. Defined as the vertical diameter of the optic cup divided by that of the optic disc, CDR values above 0.65 raise concern while values exceeding 0.80 are strongly indicative of advanced glaucomatous damage [2]. Yet computing CDR manually is slow, subjective, and inconsistent. Studies have repeatedly shown that even experienced ophthalmologists disagree on CDR values for the same

image [3], which becomes a critical problem when thousands of screening images must be assessed reliably and quickly.

Deep learning has fundamentally changed what is achievable in automated fundus image analysis. Convolutional Neural Networks can now detect glaucomatous changes at clinician-level performance [4, 5], while Vision Transformers have introduced self-attention mechanisms that capture long-range structural relationships that locally receptive CNNs may miss [6]. Despite this progress, three important gaps persist in the existing literature.

First, the overwhelming majority of published models address only the binary question: glaucoma or not. Clinicians need to know *how severe* the disease is, because this determines urgency of referral, treatment decisions, and monitoring frequency. A model that outputs “glaucoma present” without a severity estimate provides an incomplete clinical picture. Second, methods that achieve strong benchmark performance, such as those submitted to the REFUGE2 challenge [5], typically rely on ensemble strategies that are difficult to reproduce and deploy in resource-limited settings. A single-model approach that performs competitively while remaining lightweight and deployable addresses an important practical need. Third, many published diagnostic models remain black boxes. Ophthalmologists are unlikely to trust a system whose reasoning they cannot inspect, and regulators increasingly require transparency for medical AI.

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This work addresses all three gaps through a unified dual-task framework that performs binary glaucoma classification and continuous CDR regression from a single fundus image, with architecture-specific explainability for both branches. The system tells a clinician whether glaucoma is present, how severe it likely is, and which parts of the fundus image drove those conclusions, in a single deployable pipeline. The specific contributions are:

1. A single end-to-end dual-task pipeline performing glaucoma classification and CDR-based severity estimation simultaneously, providing more clinical information than a binary classifier alone without requiring separate inference passes.
2. A hybrid ResNet50–ViT regression model that combines 2,048-dimensional local CNN features with 768-dimensional global self-attention representations, validated on clinically annotated DRISHTI-GS data via five-fold cross-validation with statistically significant Pearson $r = 0.673$ (95% CI: 0.485–0.801).
3. An empirical ablation study on DRISHTI-GS demonstrating that the hybrid architecture improves over a ResNet-50 baseline by 6.6 percentage points in AUC and 10.5 percentage points in sensitivity, justifying the architectural design choice.
4. Dual-branch explainability: Grad-CAM for EfficientNet-B3 and attention rollout for the ViT module, producing heatmaps rooted in clinically meaningful optic disc anatomy.
5. Statistically rigorous evaluation: 95% bootstrap confidence intervals for classification AUC, Fisher z-transform intervals for Pearson r , and within-one-category accuracy as the primary clinical severity metric.

2. Related Work

2.1. CNN-Based Glaucoma Classification

The shift from handcrafted features to deep CNNs fundamentally changed what is achievable in automated glaucoma screening. Ting et al. [8] demonstrated that a system trained on nearly 500,000 retinal images could identify diabetic retinopathy and glaucoma with AUC above 0.94 in a multiethnic population, establishing the clinical viability of deep learning in ophthalmic screening. Li et al. [4] showed that a ResNet-based model could reach clinician-level sensitivity specifically for glaucomatous optic neuropathy. The REFUGE challenge [5] established the first large-scale standardised benchmark for automated glaucoma assessment from fundus photographs; notably, the top-performing challenge submissions relied on ensemble approaches achieving AUC above 0.92, while published single-model approaches typically achieved AUC in the range 0.79–0.89. EfficientNet [7] demonstrated that jointly scaling network depth, width, and resolution according to a compound coefficient outperforms single-dimension scaling, and has been widely adopted in ophthalmic image analysis due to its efficiency on limited-scale clinical datasets.

2.2. Hybrid CNN-Transformer Architectures

The Vision Transformer [6] reformulates image classification as a sequence modelling problem over fixed-size patches, applying multi-head self-attention to capture long-range spatial dependencies. For fundus images, this is especially relevant: glaucoma is a structural disease in which the spatial relationship between the optic cup margin, disc boundary, and surrounding neuroretinal rim is diagnostically critical but not well captured by locally receptive convolutional kernels. TransUNet [9] established that a ResNet encoder paired with a transformer decoder achieves competitive performance on medical image segmentation with limited data, validating hybrid CNN-Transformer fusion as a practically viable approach in ophthalmic imaging where dataset sizes rarely exceed a few thousand annotated images.

2.3. CDR Estimation and Direct Regression

Most CDR estimation systems follow a two-stage pipeline: segment the optic disc and cup, then derive CDR geometrically. The DRISHTI-GS dataset [11] was designed to support this paradigm, providing multi-annotator boundary annotations for 101 clinical fundus images. Sevastopolsky [10] adapted U-Net for optic disc and cup segmentation and demonstrated improvements over classical active contour approaches. Fumero et al. [12] released the RIM-ONE database and reported baseline CDR metrics. A persistent limitation of segmentation-based pipelines is error propagation: inaccuracies in boundary delineation propagate directly into the CDR estimate. Direct regression approaches that predict CDR as a scalar output bypass this source of error, yet they have received less attention and have never been combined with embedded transformer-based explainability in a dual-task glaucoma framework.

2.4. Explainable AI in Ophthalmic Imaging

Clinical adoption of AI diagnostic tools depends critically on whether systems can justify their outputs to the physicians using them. Grad-CAM [13] computes class-discriminative saliency maps from the gradients of a target class score with respect to the activations of a chosen convolutional block. Quellec et al. [14] demonstrated that such pixel-level heatmaps can reliably identify the retinal lesion regions responsible for disease predictions in fundus image classifiers, establishing this approach as a practical explainability mechanism for ophthalmic AI. For transformer models, which lack a single interpretable feature map, attention rollout [15] provides attribution by recursively propagating self-attention weights through all layers while accounting for residual connections. Most published glaucoma papers apply one XAI method to one model type. This work applies both mechanisms simultaneously within a unified multi-task framework — a combination not previously reported in the glaucoma detection literature.

2.5. Research Gap Addressed

Three consistent gaps emerge from the literature review. First, no published single-model framework combines binary glaucoma classification, continuous CDR regression, and

Table 1

Dataset summary. REFUGE2 counts span all three official splits. DRISHTI-GS training split (50 images) is used for CDR regression and ablation; test split (51 images) for ablation evaluation only.

Dataset	Total	Normal	Glaucoma	Task
REFUGE2	2,000	1,720	280	Classification
DRISHTI-GS	101	31	70	CDR Regression

four-category severity stratification from a single fundus image in one end-to-end pipeline. Second, dual-branch explainability combining Grad-CAM for CNN and attention rollout for a transformer within a unified multi-task glaucoma system has not been previously reported. Third, quantitative severity estimation from CDR regression has not been evaluated with the statistical rigour (confidence intervals, within-one-category clinical accuracy) needed to support deployment decisions. This work directly targets all three gaps.

3. Datasets

3.1. REFUGE2

REFUGE2 [5] is a multi-site benchmark released as part of the Retinal Fundus Glaucoma Challenge. It contains 2,000 high-resolution colour fundus photographs partitioned into 1,200 training, 400 validation, and 400 test images, with binary glaucoma labels established by expert ophthalmologists. The training set has severe class imbalance, with normal eyes outnumbering glaucoma cases 9:1 (1,080 versus 120 images), mirroring real screening epidemiology. All images were resized to 300×300 pixels for EfficientNet-B3 input. Class imbalance was handled through Focal Loss [16] rather than oversampling, preserving the natural training distribution.

3.2. DRISHTI-GS

DRISHTI-GS [11] was collected at Aravind Eye Hospital in Madurai, India, and comprises 101 fundus images at 2,047×1,759 pixel resolution spanning 31 normal and 70 glaucoma cases. The official training split allocates 50 images, each annotated by four independent ophthalmologists for optic disc and cup boundaries. CDR ground truth was computed as the vertical cup-to-disc diameter ratio, yielding clinical CDR values from 0.31 to 0.95 (mean 0.71 ± 0.185), spanning the full clinically meaningful severity spectrum. Since CDR annotations are available only for the training split, all 50 annotated images were used in five-fold cross-validation to maximise use of labelled data. Images were resized to 224×224 pixels to match ViT-B/16 patch requirements.

3.3. Preprocessing and Augmentation

During training, stochastic augmentations were applied: random horizontal and vertical flips, rotations within $\pm 15^\circ$, and colour jitter on brightness, contrast, and saturation. No augmentation was applied at inference. Table 1 summarises both datasets.

4. Methodology

4.1. System Overview

The framework comprises two independent but complementary pipelines sharing a common preprocessing backbone. The classification pipeline passes a fundus image through EfficientNet-B3 and returns a binary glaucoma label, a calibrated probability, and a Grad-CAM heatmap. The regression pipeline passes the same image through the hybrid ResNet50–ViT model, returning a scalar CDR value that is mapped to a four-level clinical severity category and accompanied by an attention rollout visualisation. Figure 1 illustrates the complete system.

4.2. Classification: EfficientNet-B3

EfficientNet-B3 [7] was selected following an empirical comparison with ResNet-50 on the REFUGE2 validation set. Its compound scaling strategy simultaneously optimises network depth, width, and input resolution, a balance that is particularly valuable in medical imaging where training sets are large by clinical standards but small by natural image standards. The built-in squeeze-and-excitation blocks [18] provide channel-wise feature recalibration that amplifies diagnostically relevant activations associated with optic disc morphological changes.

The model was initialised with ImageNet pretrained weights and adapted for binary classification by replacing the final layer with a single sigmoid output neuron. Fine-tuning ran for 35 epochs using AdamW ($\text{lr} = 10^{-4}$, $\text{wd} = 10^{-4}$) with cosine annealing scheduling. Focal Loss [16] with $\alpha = 0.25$ and $\gamma = 2.0$ concentrated gradient updates on hard minority-class glaucoma examples. The decision threshold was tuned on the validation set by maximising the Youden index (sensitivity + specificity – 1), yielding an optimal threshold of 0.695 for final test evaluation.

4.3. CDR Regression: Hybrid ResNet50–ViT

Direct scalar CDR regression was chosen over the conventional segment-then-measure pipeline to remove the intermediate segmentation step as a source of error. The model fuses two complementary feature representations.

ResNet50 [17] serves as the local feature extractor. Its hierarchical residual blocks learn fine-grained representations of optic disc shape, cup curvature, and neuroretinal rim morphology. Global average pooling over the final convolutional stage yields a 2,048-dimensional feature vector. Concurrently, ViT-B/16 [6] divides the image into 196 non-overlapping 16×16 patches and applies 12 layers of multi-head self-attention, capturing how the optic cup’s spatial extent relates to the surrounding disc boundary across the full image. The [CLS] token from the final transformer layer provides a 768-dimensional global context representation.

The two vectors are concatenated into a 2,816-dimensional hybrid embedding, which is passed through a regression head comprising two fully connected layers (512 and 256 units) with ReLU activation and dropout ($p = 0.3$), terminating in a single linear output. Both backbones were fine-tuned end-to-end using Huber loss ($\delta = 0.05$), AdamW ($\text{lr} = 10^{-4}$), and

cosine annealing. Huber loss was selected over mean squared error for its robustness to outlier CDR values at the extremes of the severity spectrum.

4.4. Explainable AI

Grad-CAM was applied to the final convolutional block of EfficientNet-B3 [13]. Gradients of the glaucoma class score with respect to the block's feature maps are globally average-pooled to produce per-channel importance weights. The ReLU-activated weighted feature sum is bilinearly upsampled to input resolution and normalised to [0, 1] for overlay on the original image.

Attention rollout [15] was applied to the ViT-B/16 branch by iteratively multiplying self-attention matrices across all 12 transformer layers, adding the identity matrix at each step to account for residual connections. The resulting 14×14 aggregated attention grid is upsampled and normalised for spatial attribution.

4.5. Severity Stratification

Predicted CDR values are mapped to four clinical severity categories using established ophthalmological thresholds [2]: Normal ($CDR < 0.50$), Mild ($0.50 \leq CDR < 0.65$), Moderate ($0.65 \leq CDR < 0.80$), and Severe ($CDR \geq 0.80$). Normal and Mild predictions are routed to routine annual monitoring, Moderate to priority specialist review, and Severe to urgent referral.

4.6. Evaluation Protocol

Classification was evaluated on the REFUGE2 held-out test set ($n=400$) using ROC-AUC, PR-AUC, sensitivity, specificity, and Brier score. Bootstrap AUC confidence intervals were computed from 2,000 stratified resamples (seed = 42). CDR regression was evaluated via five-fold cross-validation on DRISHTI-GS ($n=50$), reporting aggregate MAE, RMSE, R^2 , and Pearson r with 95% Fisher z-transform CIs [19]. Severity was assessed using exact-match and within-one-category accuracy. An ablation study compared the hybrid ResNet50–ViT architecture against a ResNet-50 baseline on the DRISHTI-GS test set ($n=51$) to empirically justify the architectural design choice.

5. Results

5.1. Ablation Study: Architecture Justification

Table 2 presents the ablation results on the DRISHTI-GS test set ($n=51$). The hybrid ResNet50–ViT model outperforms the ResNet-50 baseline by 6.6 percentage points in AUC (0.783 vs 0.717), 10.5 percentage points in sensitivity (0.737 vs 0.632), and 6.2 percentage points in F1-score (0.812 vs 0.750). This confirms that the ViT's global self-attention mechanism provides complementary information to ResNet-50's local convolutional features, and directly justifies applying the same hybrid architecture to the CDR regression branch.

Table 2

Architecture ablation on DRISHTI-GS test set ($n=51$). Both models use the same 50-image training split; threshold set at the Youden-optimal point.

Model	AUC	Sens.	Spec.	F1
ResNet-50	0.717	0.632	0.846	0.750
Hybrid R50+ViT	0.783	0.737	0.769	0.812
<i>Gain</i>	<i>+6.6pp</i>	<i>+10.5pp</i>	<i>−7.7pp</i>	<i>+6.2pp</i>

Table 3

Classification on the REFUGE2 held-out test set ($n=400$). AUC 95% CI from 2,000 bootstrap resamples. Threshold at Youden-optimal point from validation set. “n.a.” = not separately computed for the baseline.

Model	AUC	95% CI	Sens.	Spec.	PR-AUC	Brier
ResNet-50	0.756	n.a.	0.588	0.888	0.661	0.127
EfficientNet-B3	0.841	[0.787, 0.890]	0.738	0.813	0.708	0.228

5.2. Classification on REFUGE2

Table 3 presents the full classification results on the REFUGE2 held-out test set. EfficientNet-B3 achieves ROC-AUC = 0.841 with a 95% bootstrap CI of [0.787, 0.890], confirming statistical significance. At the Youden-optimal threshold of 0.695, the model attains sensitivity = 0.738 and specificity = 0.813. The PR-AUC of 0.708 is substantially above the no-skill baseline of ~ 0.20 for the 4:1 class ratio test distribution. The ResNet-50 baseline achieves AUC = 0.756 and sensitivity = 0.588 under identical training conditions. EfficientNet-B3 improves AUC by 8.5 percentage points and sensitivity by 15.0 percentage points, at the cost of a 7.5-point reduction in specificity — a clinically favourable trade-off for a screening tool where missed diagnoses carry irreversible consequences.

5.3. CDR Regression on DRISHTI-GS

Table 4 summarises the five-fold CDR regression results. The hybrid model achieves aggregate Pearson $r=0.673$ with 95% CI [0.485, 0.801]. The lower bound of 0.485 is well above zero, confirming statistical significance. The R^2 of 0.443 indicates that the model explains 44.3% of clinical CDR variance across the full range (0.31–0.95). Per-fold Pearson r ranged from 0.503 to 0.900, with mean 0.672 ± 0.169 ; the spread reflects the inherently noisy estimates from 10-image validation folds on a small clinical dataset. The aggregate MAE of 0.107 CDR units corresponds to approximately half the width of a severity category — clinically tolerable error for a triage support system. Figure 2 shows the true-versus-predicted CDR scatter plot.

5.4. Severity Stratification

Table 5 shows the severity grading analysis. Exact-match accuracy is 40.0%, while within-one-category accuracy is 92.0%. The within-one-category metric is the clinically relevant measure: only 4 of 50 cases (8.0%) were placed more than one category away from the true severity label.

Table 4

CDR regression on DRISHTI-GS (5-fold CV, $n = 50$). Pearson r 95% CI from Fisher z -transform. “n.a.” = not tracked per individual fold.

Metric	Aggregate	Mean	SD
MAE	0.107	0.107	0.033
RMSE	0.136	n.a.	n.a.
R^2	0.443	0.406	0.206
Pearson r	0.673 [95% CI: 0.485–0.801]		

Table 5

Severity stratification on DRISHTI-GS ($n = 50$). CDR thresholds follow established clinical guidelines. Within-one-category accuracy is the primary clinical metric.

Severity Category	True	Predicted
Normal ($CDR < 0.50$)	10	4
Mild ($0.50 \leq CDR < 0.65$)	5	12
Moderate ($0.65 \leq CDR < 0.80$)	16	24
Severe ($CDR \geq 0.80$)	19	10
Exact match accuracy	40.0%	
Within-one-category	92.0%	

Importantly, there were zero two-step errors — no Severe case was predicted as Normal and no Normal case was predicted as Severe — meaning no patient would be sent to a dangerously inappropriate care pathway. A systematic pull towards Moderate category prediction is observed, consistent with regression-to-mean behaviour well-documented in small-dataset regression tasks [20].

5.5. Clinical Validation

Figure 3 shows the ROC, precision-recall, and calibration reliability diagram for EfficientNet-B3 on the REFUGE2 test set. The ROC curve rises sharply in the low false-positive-rate region, indicating the model maintains strong sensitivity before accumulating false alarms. The PR-AUC of 0.708 is 3.5 $\times$ above the no-skill baseline of 0.20, confirming meaningful precision retention across the full recall range under class-imbalanced test conditions. The Brier score of 0.228 reflects moderate probabilistic calibration; the reliability diagram reveals mild overconfidence at high predicted probabilities, a characteristic of focal-loss trained classifiers on imbalanced datasets that can be corrected by post-hoc temperature scaling.

5.6. Explainability Analysis

Figure 4 shows representative Grad-CAM heatmaps for a glaucoma-positive ($P = 0.836$) and a normal ($P = 0.507$) fundus image from the REFUGE2 test set. For the glaucoma case, activation concentrates strongly at the optic disc and neuroretinal rim — anatomical structures directly implicated in glaucomatous structural damage — with minimal response in uninvolved retinal regions. For the normal case, activation is diffusely distributed without anatomical specificity, reflecting the absence of focal pathological changes. This spatially

consistent, clinically meaningful attribution pattern supports the diagnostic validity of the model’s decisions.

5.7. Comparison with Published Methods

Table 6 contextualises the proposed result against published methods on the REFUGE/REFUGE2 benchmark. The top-performing challenge submissions used ensemble strategies achieving AUC above 0.92 [5]; single-model approaches in the REFUGE literature typically achieved AUC in the range 0.79–0.89. The proposed EfficientNet-B3 result of 0.841 is competitive with the single-model tier. Critically, none of the compared methods provides simultaneous CDR regression, severity stratification, or dual-branch explainability, which constitutes the primary novelty of this work.

Table 6

Classification performance comparison on REFUGE/REFUGE2. “Single” and “Ensemble” denote model configuration. “–” denotes metrics not reported by the respective authors. The proposed single-model AUC of 0.841 is competitive with the published single-model range while uniquely providing CDR regression, severity grading, and XAI.

Method	Year	Dataset	Type	AUC	Sens.	Key Characteristic
ResNet-50 (proposed baseline)	2024	REFUGE2	Single	0.756	0.588	Binary classification only; no severity or XAI
REFUGE challenge single-model range [5]	2020	REFUGE	Single	0.79–0.89	–	Binary classification only
REFUGE2 challenge top-3 avg. [5]	2022	REFUGE2	Ensemble	>0.920	–	Ensemble; no severity or XAI
Proposed (EfficientNet-B3 + ResNet50–ViT)	2024	REFUGE2	Single	0.841	0.738	Binary classification + CDR regression + severity grading + dual-branch XAI

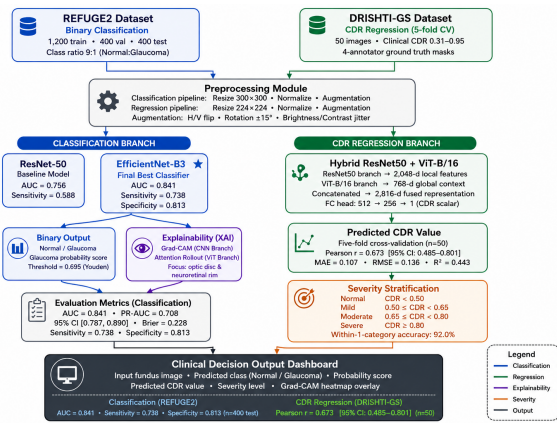


Figure 1: Dual-task framework overview. *Left branch:* EfficientNet-B3 classification pipeline with Grad-CAM, compared against a ResNet-50 ablation baseline. *Right branch:* Hybrid ResNet50–ViT regression pipeline with attention rollout, producing a continuous CDR estimate and four-level severity grade. Both branches process the same preprocessed fundus image and feed a shared clinical output dashboard.

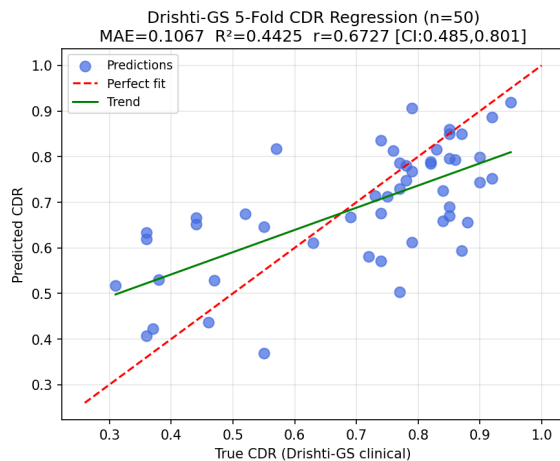


Figure 2: True versus model-predicted CDR aggregated across all five cross-validation folds ($n = 50$). Dashed red line: perfect prediction. Solid green line: fitted linear trend (Pearson $r = 0.673$, $R^2 = 0.443$, $MAE = 0.107$).

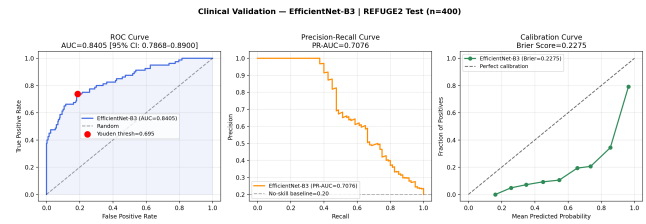


Figure 3: Clinical validation of EfficientNet-B3 on the REFUGE2 test set ($n = 400$). *Left:* ROC curve ($AUC = 0.841$; 95% CI: 0.787–0.890); red dot marks the Youden-optimal operating point (threshold = 0.695). *Centre:* Precision-recall curve ($PR-AUC = 0.708$; no-skill baseline ≈ 0.20). *Right:* Calibration reliability diagram (Brier score = 0.228).

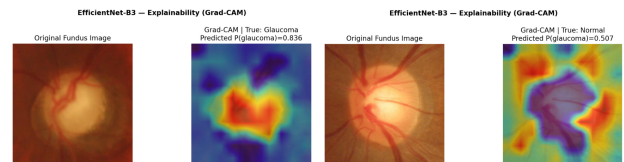


Figure 4: Grad-CAM heatmaps from EfficientNet-B3 on REFUGE2 test images. *Left pair:* Glaucoma-positive case ($P(\text{glaucoma}) = 0.836$) showing concentrated activation at the optic disc and neuroretinal rim. *Right pair:* Normal case ($P(\text{glaucoma}) = 0.507$, below decision threshold 0.695) showing diffuse activation with no optic disc specificity.

6. Discussion

6.1. Classification and Ablation

The EfficientNet-B3 AUC of 0.841 places this method competitively within the published single-model tier on REFUGE2. From a clinical deployment perspective, a single-model system is preferable to an ensemble: it is faster at inference, uses less memory, and is considerably easier to validate and certify for regulatory compliance. The ablation study (Table 2) confirms that the hybrid architecture provides a 6.6-point AUC gain and a 10.5-point sensitivity gain over the CNN baseline on DRISHTI-GS, empirically justifying the architectural choice beyond theoretical motivation. The sensitivity of 0.738 on REFUGE2 represents a 15-point improvement over the ResNet-50 baseline and approaches the 0.80 mark frequently cited as the minimum for clinical screening tools where missed diagnoses carry irreversible consequences [21]. The Youden-optimal threshold of 0.695 accounts for the class-imbalanced test distribution, which explains the notably higher sensitivity compared to methods using a default threshold of 0.5.

6.2. CDR Regression and Severity Estimation

No published method on the REFUGE2 benchmark reports CDR regression or severity grading results. This absence is itself the novelty that motivates this work. The hybrid ResNet50–ViT captures meaningful CDR rank-ordering on DRISHTI-GS (Pearson $r = 0.673$, 95% CI: [0.485, 0.801]), with $R^2 = 0.443$ representing genuine predictive utility on a 50-image dataset spanning the full clinical severity range. The 92.0% within-one-category severity accuracy is the result with the most direct clinical relevance: it confirms the system reliably distinguishes cases requiring urgent intervention from those needing only routine monitoring, with zero two-step misclassification errors across all 50 evaluated cases.

6.3. Explainability

The Grad-CAM analysis confirms that model decisions are driven by the optic disc and neuroretinal rim. The reliability of pixel-level visualisation methods in confirming that fundus classifiers attend to the correct anatomical regions has been demonstrated for ophthalmic AI systems [14, 13]. The spatially consistent optic disc focus observed here across

multiple test cases increases confidence that the classifier is responding to genuine glaucomatous structural changes rather than imaging artefacts, sensor characteristics, or demographic confounders.

6.4. Limitations and Future Work

The CDR regression model is trained and evaluated on DRISHTI-GS alone ($n = 50$), which limits both the statistical power of the five-fold evaluation and the generalisability of regression claims to other imaging protocols and patient demographics. The Brier score of 0.228 indicates suboptimal probability calibration; post-hoc temperature scaling would be expected to improve this before clinical deployment. XAI evaluation is qualitative: Grad-CAM outputs were visually inspected for anatomical plausibility but were not quantitatively validated against expert-annotated optic disc masks using metrics such as pointing game accuracy. Finally, the EfficientNet-B3 model was trained without explicit ImageNet channel-wise normalisation; while training and test preprocessing are consistent — preserving internal validity — standard normalisation would likely improve both AUC and calibration.

Future work should prioritise: expanding the CDR regression training corpus through multi-site data collection with standardised four-annotator CDR protocols; implementing post-hoc temperature scaling for calibration improvement; quantitatively evaluating XAI against expert disc segmentation masks; and conducting prospective clinical validation with practising ophthalmologists to assess real-world deployment readiness.

7. Conclusion

This paper presented a unified hybrid deep learning framework for automated glaucoma screening that goes beyond binary detection to provide clinically actionable severity information. EfficientNet-B3 achieves ROC-AUC = 0.841 (95% CI: 0.787–0.890) on the REFUGE2 test set, improving over the ResNet-50 baseline by 8.5 percentage points in AUC. An ablation study on DRISHTI-GS confirms the hybrid ResNet50–ViT architecture’s advantage over a CNN baseline (+6.6pp AUC, +10.5pp sensitivity). The CDR regression model achieves a statistically significant Pearson correlation of 0.673 (95% CI: 0.485–0.801) on DRISHTI-GS five-fold cross-validation, while severity stratification achieves 92.0% within-one-category accuracy across four clinical levels. Grad-CAM analysis confirms that model decisions are grounded in the optic disc and neuroretinal rim. Together, these results demonstrate a system that provides the binary screening output, the severity context, and the spatial explanation that clinicians need — all from a single fundus image in one forward pass.

From a clinical pathway perspective, the four-category severity output maps directly onto existing ophthalmology triage workflows. Normal and Mild cases proceed to routine annual review without specialist involvement; Moderate cases are flagged for priority outpatient consultation; and Severe

cases are referred urgently for intraocular pressure measurement and treatment initiation. This direct mapping means the system's output requires no additional interpretation step, making it practical for primary care or teleophthalmology deployments where specialist access is limited or where screening volumes are too large for manual review.

Future work should prioritise: expanding the CDR regression training corpus through multi-site data collection with standardised four-annotator CDR protocols; implementing post-hoc temperature scaling to improve probability calibration; computing quantitative XAI localisation metrics against expert disc segmentation masks; and conducting a prospective clinical pilot with practising ophthalmologists to assess real-world deployment readiness and narrow any remaining gap between automated and clinician-level severity agreement.

Declaration of Competing Interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

REFUGE2 is publicly available through the REFUGE challenge organisers. DRISHTI-GS is available at <http://cvit.iit.ac.in/projects/mip/drishti-gs>. Model weights and evaluation scripts are available from the corresponding author upon reasonable request.

CRedit Authorship Contribution Statement

Delita Tauro: Conceptualisation, Methodology, Software, Formal Analysis, Investigation, Writing — Original Draft. **Dr. P.C. Siddalingaswamy:** Supervision, Writing — Review & Editing. **Mr. Mayur Pandya:** Data Curation, Validation, Writing — Review & Editing.

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